Impact of Ocean Acidification on Marine Calcifiers: A Meta-Analysis

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Abstract

Marine organisms with calcium carbonate skeletons are vulnerable to chemical shifts in seawater carbonate chemistry.

High atmospheric CO2 concentrations significantly reduce calcification rates in marine calcifiers.

Rising ocean temperatures directly accelerate coral bleaching severity across tropical reefs.

Results and Ecosystem Impact

Across 42 controlled flume experiments, net skeletal growth declined by an average of 22 percent.

Ocean acidification reduces shell thickness in Antarctic pteropods.

These observations demonstrate severe systemic vulnerabilities in pelagic ecosystems.